Universidad Carlos III de Madrid



Computational fluid dynamics

Lid Driven Cavity Project

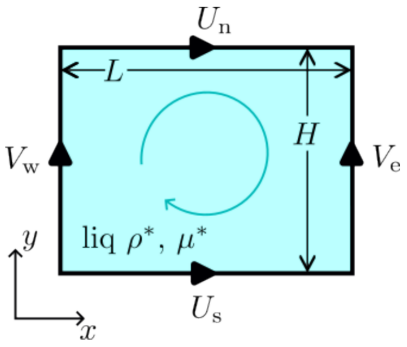
Bachelor in Industrial Technologies Engineering

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1 Problem statement

The objective is to calculate the laminar, steady flow of a constant properties fluid in a cavity driven by the motion of one or more of its walls. The cavity dimensions are L^*xH^* , as shown in the sketch. We identify the cavity walls by a compass or cardinal notation (north, south, east and west) and denote the velocity of each wall with the corresponding subscript. Note that each wall can only move parallel to itself. The fluid has density ρ^* and viscosity μ^* . In this problem statement dimensional variables are indicated with an asterisk as a superscript. Variables with no asterisk are dimensionless.



2 Mathematical model

2.1 Definition of non-dimensional variables

To simplify the calculations and the code, dimensionless variables are declared as such :

$$L = 1, H = \frac{H^*}{L^*}, x = \frac{x^*}{L^*}, y = \frac{y^*}{L^*}, t = \frac{t^* v^*_{ref}}{L^*}, v_x = \frac{v^*_x}{v^*_{ref}}, v_y = \frac{v^*_y}{v^*_{ref}}, p = \frac{p^*}{\rho^* v^{*2}_{ref}}, Re = \frac{\rho^* v^*_{ref} L^*}{\mu^*}$$

$$U_n = \frac{U^*_n}{v^*_{ref}}, U_s = \frac{U^*_s}{v^*_{ref}}, U_e = \frac{U^*_e}{v^*_{ref}}, U_w = \frac{U^*_w}{v^*_{ref}}$$

with $v^*_{ref} = max(U^*_n, U^*_s, U^*_e, U^*_w)$.

2.2 Equations

It is an incompressible flow with constant physical properties. The movement is twodimensional, planar and the flow regime is laminar. Even though the flow is steady, we will write the equations retaining the time derivative because we will use a pseudotransient method to solve the problem. The equations that govern the flow field are the continuity equation and the momentum equation. In Cartesian coordinates,

$$\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} = 0 \tag{1}$$

$$\frac{\partial v_x}{\partial t} + v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_y}{\partial y} = -\frac{\partial p}{\partial x} + \frac{1}{Re}\left(\frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_x}{\partial y^2}\right) \tag{2}$$

$$\frac{\partial v_y}{\partial t} + v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_y}{\partial y} = -\frac{\partial p}{\partial y} + \frac{1}{Re}\left(\frac{\partial^2 v_y}{\partial x^2} + \frac{\partial^2 v_y}{\partial y^2}\right) \tag{3}$$

Solving equations (1)-(3), the velocity $\vec{v}(x, y)$ and pressure $p(x, y)$ could be determined. However, in this problem the alternative formulation in Helmholtz variables (vorticity and stream function) is very advantageous, since it does not include the pressure and reduces by one the number of variables and of equations. For a planar

two-dimensional flow, the vorticity has only one component and thus we can work with its magnitude as scalar field, $\omega(x, y)$,

$$\omega = \frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \tag{4}$$

The stream function ψ is defined as

$$v_x = \frac{\partial \psi}{\partial x} \qquad v_y = -\frac{\partial \psi}{\partial x} \tag{5}$$

The system of equations for ω and ψ are:

$$\frac{\partial \omega}{\partial t} + \frac{\partial \omega v_x}{\partial x} + \frac{\partial \omega v_y}{\partial y} = \frac{1}{Re}(\frac{\partial^2 \omega}{\partial x^2} + \frac{\partial^2 \omega}{\partial y^2}) \tag{6}$$

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = -\omega \tag{7}$$

The first is a transport equation for the vorticity that is obtained by taking the curl of the momentum equation, that is, taking the derivative with respect to y of the equation (2) and subtracting the derivative with respect to x of the equation (3). The second equation is obtained by substituting the expressions (5) into the definition of vorticity (4). The two equations constitute a complete system that allows calculating the distributions of ψ and ω in the fluid field. From these results the velocity field can be obtained applying (5). Once the velocity field is known in the flow domain, the pressure field may be obtained by solving a Poisson equation

$$\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} = (\frac{\partial u}{\partial x})^2 + 2\frac{\partial u}{\partial y}\frac{\partial v}{\partial x} + (\frac{\partial v}{\partial y})^2 \tag{8}$$

derived by taking the divergence of the momentum equation.

2.3 Flow domain

The flow domain in this problem is clearly defined

$$0 \leq x \leq 1$$

$$0 \leq y \leq H$$

2.4 Boundary conditions

2.4.1 Boundary conditions for the velocity

The no slip conditions on each wall gives the following boundary conditions for the velocity components

north wall, $(0 < x < L, y = H)$: $\quad v_x = U_n \quad v_y = 0$

south wall, $(0 < x < L, y = 0)$: $\quad v_x = U_s \quad v_y = 0$

east wall, $(x = L, 0 < y < H)$: $\quad v_x = 0 \quad v_y = U_e$

east wall, $(x = 0, 0 < y < H)$: $\quad v_x = 0 \quad v_y = U_w$

2.4.2 Boundary conditions for the streamfunction

Non-permeable walls (with $\vec{v} \cdot \vec{n} = 0$) are always streamlines, thus the streamfunction is constant along the walls. In the cavity of this problem, the four walls are actually the same streamline, with the same value of the streamfunction ψ. Setting the reference value for the streamfunction $\psi(x = 0, y = 0) = 0$, the streamfunction on all walls is null.

2.4.3 Boundary conditions for the vorticty

In non-permeable walls, the no-slip condition establishes that $\vec{v} = \vec{v_{wall}}$. In the cavity problem, the walls may have only tangential velocity. In a general formulation for all the cavity walls, let n and s be the coordinates normal to the wall (pointing to the fluid) and along the wall (such that (n, s, z) is a right-handed coordinate system). Say

$v_n = \frac{\partial \psi}{\partial s}$ and $v_s = -\frac{\partial \psi}{\partial n}$

Then, since $v_{n,wall} = 0$

$\frac{\partial \psi}{\partial s} = 0$ and also $\frac{\partial^2 \psi}{\partial s^2} = 0$

Thus, equation (7) evaluated on the walls reduces to

$$\omega_{wall} = -\frac{\partial^2 \psi}{\partial n^2} \tag{9}$$

On the other hand,

$$v_{s,wall} = -\frac{\partial \psi}{\partial n} \tag{10}$$

Vorticity on walls is calculated from (9) but reinforcing (10).
Writing the relations for each wall

$$\begin{aligned}
&\text{north wall, } (0 < x < L, y = H): && \omega_n = -\tfrac{\partial^2 \psi}{\partial y^2} && U_n = \tfrac{\partial \psi}{\partial y} \\
&\text{south wall, } (0 < x < L, y = 0): && \omega_s = -\tfrac{\partial^2 \psi}{\partial y^2} && U_s = \tfrac{\partial \psi}{\partial y} \\
&\text{east wall, } (x = L, 0 < y < H): && \omega_e = -\tfrac{\partial^2 \psi}{\partial x^2} && U_e = \tfrac{\partial \psi}{\partial x} \\
&\text{east wall, } (x = 0, 0 < y < H): && \omega_w = -\tfrac{\partial^2 \psi}{\partial x^2} && U_n = \tfrac{\partial \psi}{\partial x}
\end{aligned}$$

2.5 Discretization

2.5.1 Computational grid

The first step of discretization is to create a computational grid, for every case a number of points N will be set, $N = M$ with N the amount of samples on the x and y axis. This makes the discretization non-uniform and therefore some simplifications of the scheme cannot be made. However this will make the calculations for a cavity with a non unitary aspect ratio more accurate :

$$\Delta x = \frac{1}{N-1} \qquad \Delta y = \frac{H}{N-1}$$

Δt will be set afterwards to stabilize the schemes

2.5.2 Numerical scheme

Our system of equation is composed of equations (5) (6) and (7),for this problem we will be only using explicit schemes which are simple but tend to have a consequent error with small grids.For the calculation of the velocities (5) we will be using centered differences for v_x and v_y,

$$v^t_{y\,i,j} = -\frac{\psi^t_{i+1,j} - \psi^t_{i-1,j}}{2\Delta x} \qquad v^t_{x\,i,j} = \frac{\psi^t_{i,j+1} - \psi^t_{i,j-1}}{2\Delta y}$$

for equation (6) we will be using an explicit scheme with forward differences in time t and centered differences in x and y

$$\frac{\omega^{t+1}_{i,j} - \omega^t_{i,j}}{\Delta t} + \frac{(v^t_{x\,i+1,j}\omega^t_{i+1,j}) - (v^t_{x\,i-1,j}\omega^t_{i-1,j})}{2\Delta x} + \frac{(v^t_{y\,i,j+1}\omega^t_{i,j+1}) - (v^t_{y\,i,j-1}\omega^t_{i,j-1})}{2\Delta y} =$$

$$\frac{1}{Re}(\frac{\omega^t_{i+1,j}+\omega^t_{i-1,j}-2\omega^t_{i,j}}{\Delta x^2}+\frac{\omega^t_{i,j+1}+\omega^t_{i,j-1}-2\omega^t_{i,j}}{\Delta y^2})$$

for equation (7) we will be using centered differences in x and y

$$\frac{\psi^t_{i+1,j}+\psi^t_{i-1,j}-2\psi^t_{i,j}}{\Delta x^2}+\frac{\psi^t_{i,j+1}+\psi^t_{i,j-1}-2\psi^t_{i,j}}{\Delta y^2}=\omega^t_{i,j}$$

To solve for $\psi^t_{i,j}$ we will be using the Successive over-relaxation method to solve the Laplace equation, with $\beta=\frac{\Delta x^2}{\Delta y^2}$ and $0<f<2$

$$\psi_{i,j}=f\frac{1}{2(1+\beta^2)}(\psi_{i+1,j}+\psi_{i-1,j}+\beta^2(\psi_{i,j+1}+\psi_{i,j-1})+\Delta x^2\omega_{i,j})+(1-f)\psi^{t-1}_{i,j}$$

To speed up convergence we will be using the optimal relaxation factor for the Laplace equation

$$f_{opt}=\frac{2}{(1+\sqrt{1-\rho^2})}$$

with

$$\rho=0.25(cos(\frac{\pi}{N-1})+cos(\frac{\pi}{M-1}))$$

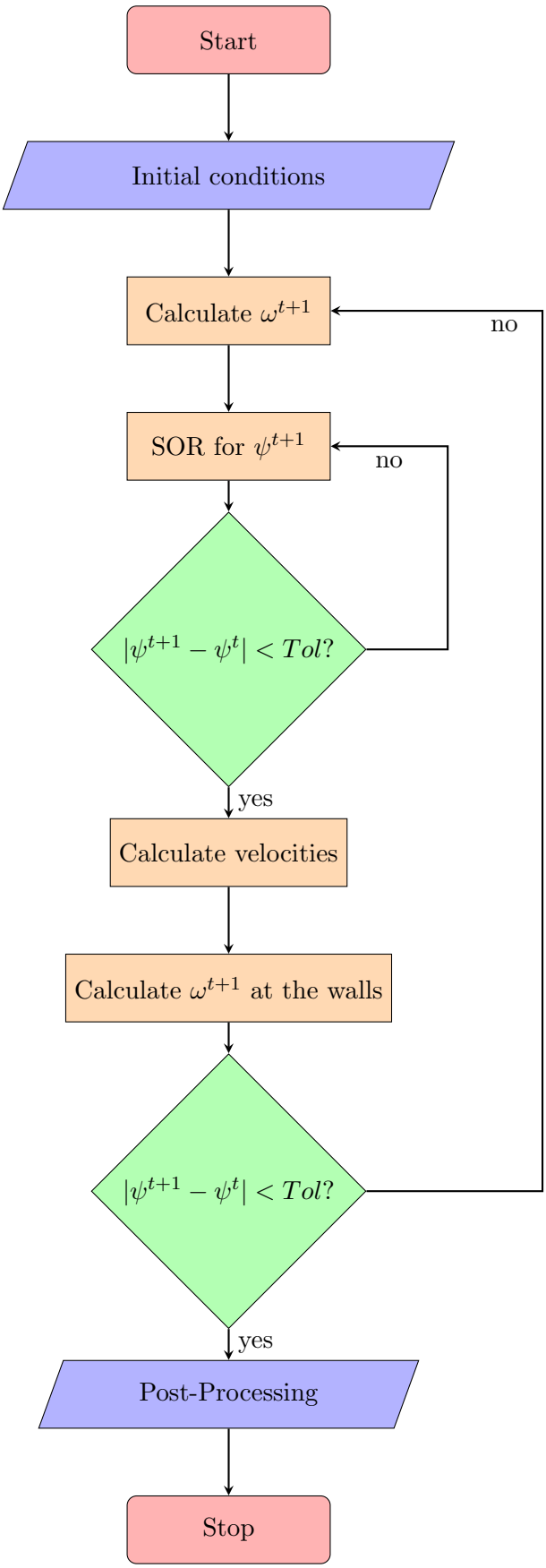
2.5.3 Calculating ω at the walls

For the numerical implementation of the vorticity boundary conditions, we are gonna use a second order approximation derived from the taylor series of $\psi(n_{wall}+\Delta n,s)$,

$$\omega_{wall}(s)=\frac{2}{\Delta n^2}(7\psi_{wall}-8\psi(n_{wall}+\Delta n,s)+\psi(n_{wall}+2\Delta n,s)-6\Delta n v_{s,wall})+O(\Delta n^2)$$

3 Method of resolution

The code for solving the equations (5) (6) and (7) will be implemented into Matlab and will follow the flow chart below. The initial conditions for this problem will all be zeros matrices except for the velocity for which the boundary conditions will be applied directly. The calculations will stop when the streamfunction reaches a "steady" state, characterized by its change at each time step


Start
Initial conditions
Calculate ω^{t+1}
no
SOR for ψ^{t+1}
no
$|\psi^{t+1} - \psi^t| < Tol?$
yes
Calculate velocities
Calculate ω^{t+1} at the walls
$|\psi^{t+1} - \psi^t| < Tol?$
yes
Post-Processing
Stop

4 Code validation and grid independency study on canonical case

4.1 grid independency study

For the grid independency study of the we will be comparing the different results to the reference [1]. the parameters used for the calculations will be : $H = 1 \quad U_e = 0 \quad U_w = 0 \quad U_s = 0 \quad U_n = -1 \quad Re = 1000$, different values of N will be used to compare the accuracy of our scheme to the reference.
the results we will be comparing are : the intensity and the position of the primary vortex and the value of the vorticity at the primary vortex, the primary vortex will be characterized by the maximum of the streamfunction and the vortex center will correspond to the coordinates of this maximum. Additionnally we will be comparing the lower left secondary vortex.

4.1.1 comparison of the primary vortex

Intensities of the primary vortex at Re=1000; (x,y) refers to the center of the primary vortex					
Reference	Grid size (N)	ψ	ω	x	y
Present	24	0.1000045	1.7907	0.500	0.5833
Present	48	0.1130354	1.9700	0.4792	0.5833
Present	64	0.1154715	2.0095	0.4688	0.5781
Present	96	0.1173174	2.0401	0.4688	0.5729
Present	128	0.1176657	2.0597	0.4766	0.5703
Present	160	0.1183217	2.0570	0.4750	0.5688
Present	201	0.1186187	2.0640	0.4726	0.5672
Present	500	0.1188672	2.0666	0.4700	0.5660
Reference	48	0.1189249	2.067396	0.4692	0.5652
Reference	64	0.1189365	2.068016	0.4692	0.5652
Reference	96	0.1189366	2.067750	0.4692	0.5652
Reference	128	0.1189366	2.067750	0.4692	0.5652
Reference	160	0.1189366	2.067753	0.4692	0.5652

4.1.2 comparison of the secondary vortex

Intensities of the lower left secondary vortex at Re=1000; (x,y) refers to the center of the vortex					
Reference	Grid size (N)	ψ	ω	x	y
Present	24	-2.483352 x 10^{-3}	-1.262304	0.1667	0.1667
Present	48	-1.831532 x 10^{-3}	-1.289357	0.1667	0.1250
Present	64	-1.792432 x 10^{-3}	-1.236631	0.1563	0.1250
Present	96	-1.754190 x 10^{-3}	-1.185061	0.1458	0.1250
Present	128	-1.414338 x 10^{-3}	-0.546442	0.1094	0.1094
Present	160	-1.737109 x 10^{-3}	-1.071302	0.1375	0.1188
Present	201	-1.734928 x 10^{-3}	-1.074285	0.1393	0.1144
Present	500	-1.730430 x 10^{-3}	-1.119690	0.1380	0.1140
Reference	48	-1.728963 x 10^{-3}	-1.109440	0.1360	0.1118
Reference	64	-1.728963 x 10^{-3}	-1.109714	0.1360	0.1118
Reference	96	-1.729716 x 10^{-3}	-1.109772	0.1360	0.1118
Reference	128	-1.729717 x 10^{-3}	-1.109794	0.1360	0.1118
Reference	160	-1.729717 x 10^{-3}	-1.109789	0.1360	0.1118

4.2 Code validation

for the code validation, the figures of the streamlines and the iso vorticity will be compared to the figures of the reference [1] (figures 1 and 2).for the figures to be accurately compared the contour plot will be drawn using the same values of streamlines and iso vorticity they can be found on tables 7 and 8 of the reference [1].Additionally, the vorticities on the moving wall (y=1) will be compared.

4.2.1 Figures comparison

As it can be seen below the main differences between the figures are the secondary vortices of the streamfunction.

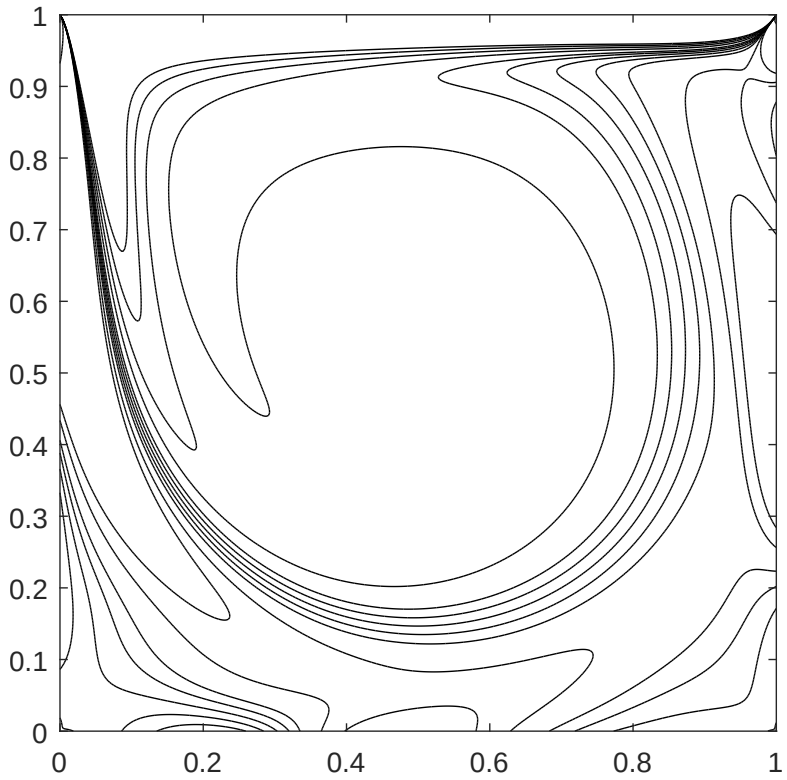


Figure 1: Iso-vorticity lines for Re=1000 N=500

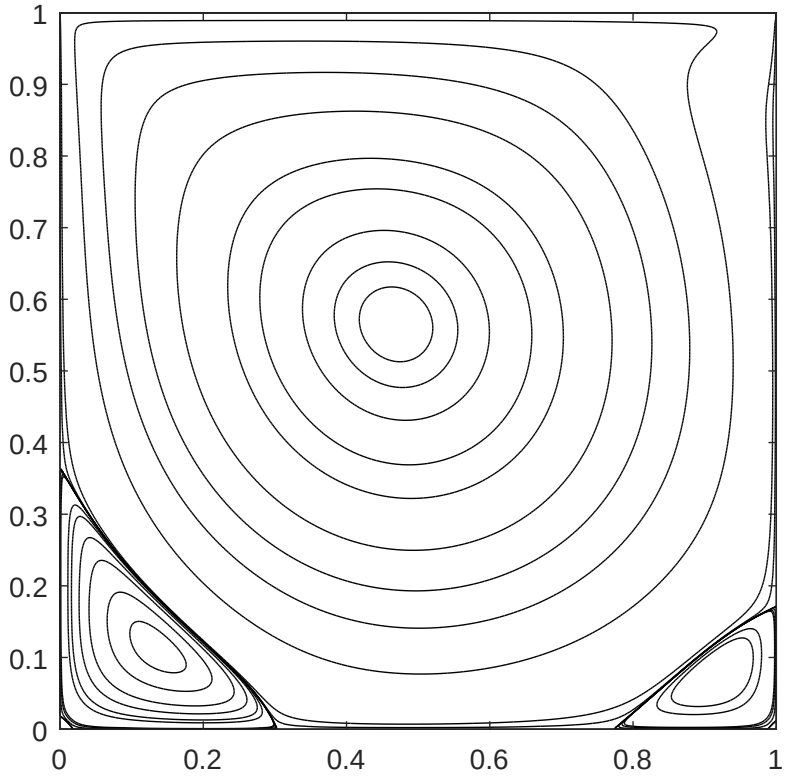
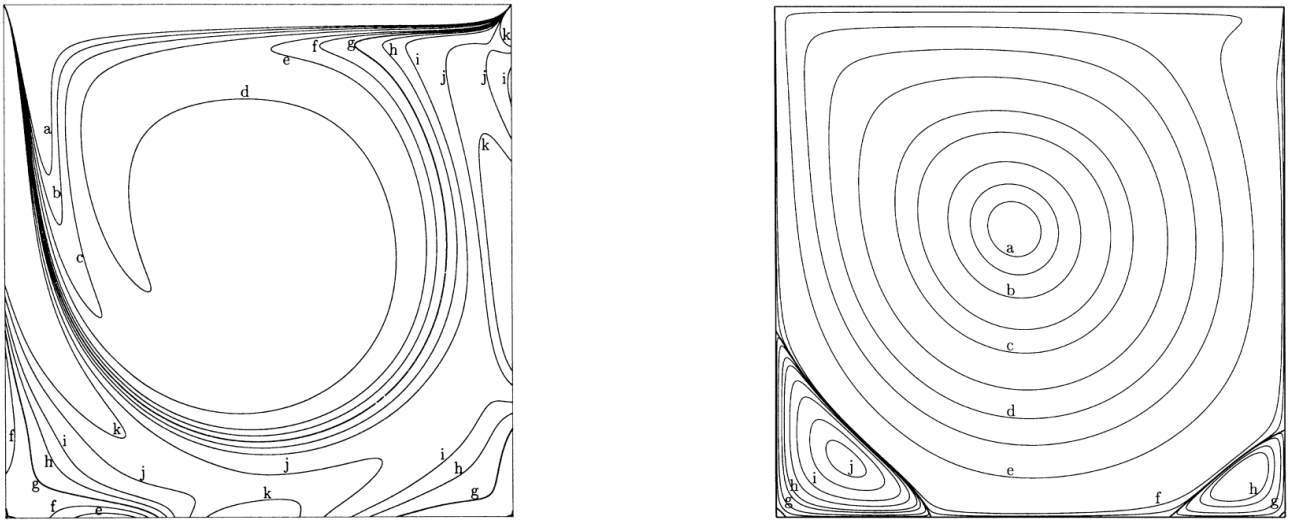


Figure 2: Streamlines for Re=1000 N=500



(a) Iso-vorticity lines.

(b) Streamlines

Figure 3: Figures 1 and 2 of reference [1].

4.2.2 Vorticity on the moving wall

Vorticity values at the moving wall (y=1) at Re=1000, N=500		
x	ω, ref [1]	ω
1.0000	—	—
0.9375	72.63795	72.99865
0.8750	50.50730	50.70168
0.8125	40.21384	40.11238
0.7500	32.01224	31.96902
0.6875	25.19252	25.19624
0.6250	20.06667	20.09914
0.5625	16.66992	16.62813
0.5000	14.75342	14.74736
0.4375	13.98148	13.99155
0.3750	14.05334	14.06615
0.3125	14.75229	14.79048
0.2500	16.02919	16.07504
0.1875	18.35666	18.42093
0.1250	24.05671	24.17262
0.0625	42.70770	44.09956
0.0000	—	—

4.3 Conclusion

It can be seen that the explicit scheme is very dependant of its discretization, with solutions using grids with a size N inferior to 100 having a substantial error, however it must be noted that this scheme converges relatively fast despite not being very accurate.

5 Convergence information and computational cost

5.1 Convergence information

The explict schemes are known to be conditionally stable with the scheme for vorticity being the limiting factor in the code. Since the Reynolds number, Δx , Δy, are fixed the only parameter that can be changed is the time step Δt.The condition over Δt, taken from reference [3] to make the explicit scheme with centered differences converge is :

$$\Delta t = min(\frac{2}{Re}, \frac{Re(\Delta x^2 + \Delta y^2)}{2})$$

To characterize the steadiness of the equation, we will be comparing ψ^{t+1} and ψ^t. If their difference is more than $1 * 10^{-10}$ solution is deemed unsteady.

The error of the solution is of the order of Δt , Δx^2 and Δy^2 which makes the solution very inaccurate for smaller grid sizes.

5.2 Computational cost

The computer used for the calculations is an MSI gf65 thin 10ue with an i5-10500h processor, 16Gb of ram and an NVMe ssd.
It must be noted that the cpu time might not be very accurate since the perfomance of a laptop is very dependant on its battery

Computational cost for every grid size at Re=1000 and N=500		
Grid size [n]	CPU time (s)	iterations
24	0.281402	4400.0
48	1.165931	11400
64	2.890724	16700
96	9.876788	28800
128	30.80641	43000
160	59.74731	59600
201	158.4332	84900
500	6510.316	578623

6 Result for particular case

For my particular case i was assigned to compute the flow for $Re \in \{1, 10, 100, 100\}$ with the following parameters:
$U_s = 0 \quad V_w = 0 \quad U_n = 1 \quad V_e = 1 \quad H = 0.5$

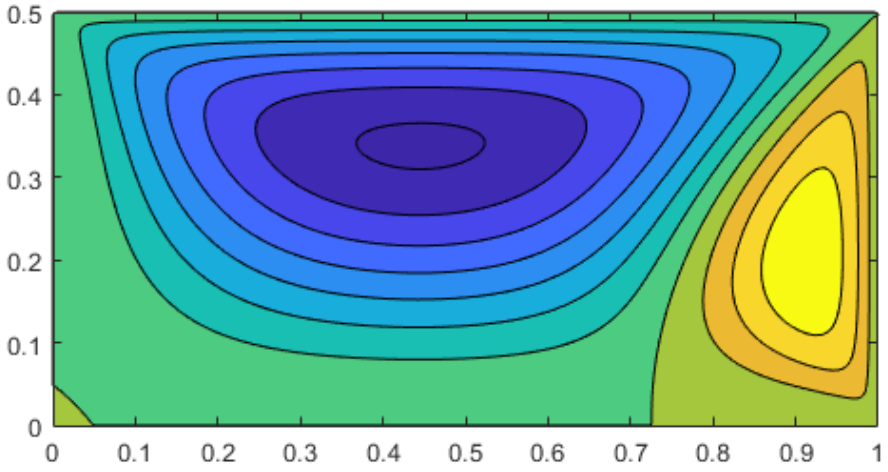


Figure 4: Streamlines for Re=1 N=200

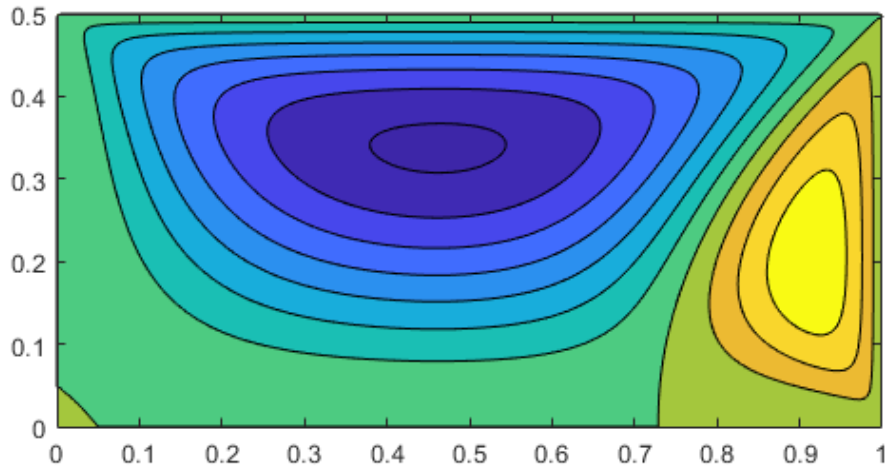


Figure 5: Streamlines for Re=10 N=200

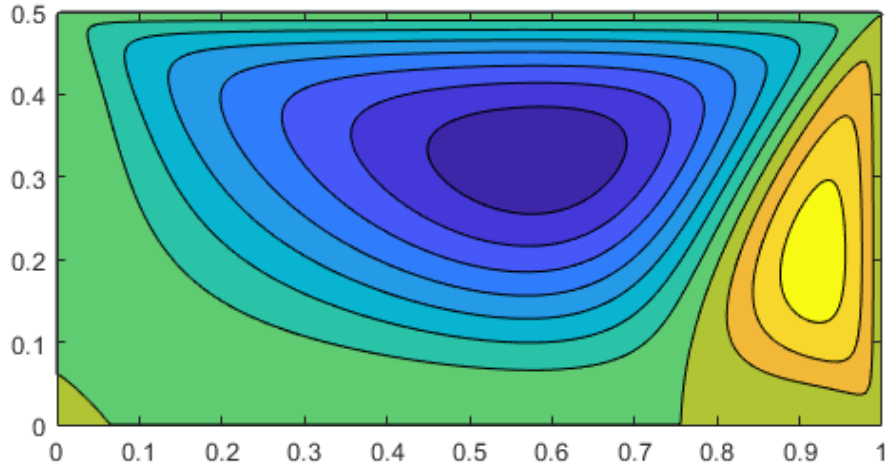


Figure 6: Streamlines for Re=100 N=200

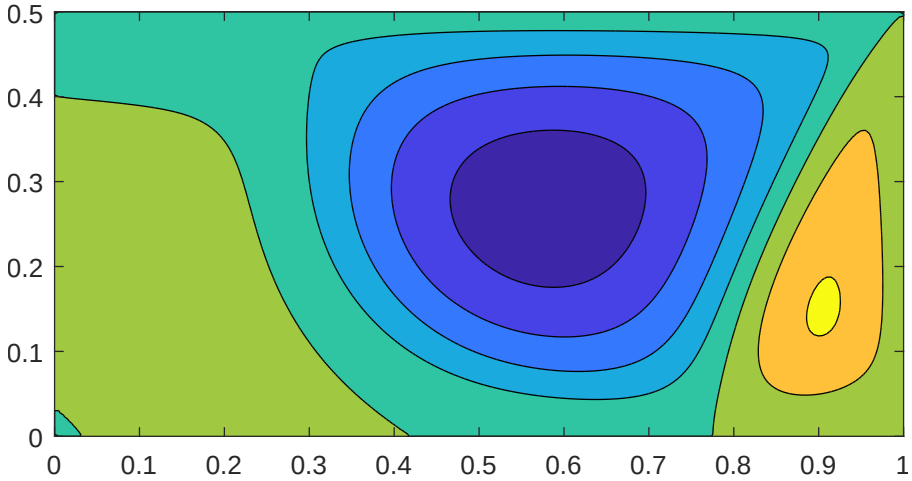


Figure 7: Streamlines for Re=1000 N=200

7 MatLab Code

```
clc;
clear;
close all;

    %% Physical Parameters
Re=1000;
H=1;
Us=0;
Un=-1;
Ve=0;
Vw=0;
L=1;

    %% Numerical parameters
Ng=201;
dx=1/(Ng-1);
dy=H/(Ng-1);

dt=min(2/Re,(Re*(dx^2+dy^2))/2);
Beta2=(dx/dy)^2;

kmax=1000000;

X=0:dx:L;
Y=0:dy:H;
N=length(X);
M=length(Y);
tolw=1e-10;
tolpsi=1e-4;

rho=0.25*(cos(pi/(N-1))+cos(pi/M-1));
f=2/(1+sqrt(1-rho^2));
    %% initial conditions

w=zeros(N,M);
psi=zeros(N,M);
pressure=zeros(N,M);
velocity_y=zeros(N,M);
velocity_x=zeros(N,M);

velocity_x(:,1)=Us;
velocity_x(:,M)=Un;
velocity_y(1,:)=Vw;
velocity_y(N,:)=Ve;

Err=1;
k=0;

tic
%% loop
while (Err > tolw && k<kmax) | k < 100 % the or k<100 is because of psi not
   changing the first iterations
    k=k+1;
    % wold = w;
    psi_old=psi;
    % calculation of vorticity outside of walls
    for i=2:N-1
        for j=2:M-1
```

```
        %explicit quasi-linear
        % w(i,j)= wold(i,j) + (dt/(2*dx))*(-velocity_x(i,j)*(wold(i+1,j)
            -wold(i-1,j)))...
        %                   +(dt/(2*dy))*(-velocity_y(i,j)*(wold(i,j+1)-
            wold(i,j-1)))...
        %                   + (dt/(Re*dx^2))*(wold(i+1,j)+wold(i-1,j)-2*
            wold(i,j)) + (dt/(Re*dy^2))*(wold(i,j+1)+wold(i,j-1)-2*wold(i
            ,j));

        %explicit upwind scheme
        % % w(i,j)= wold(i,j) - ((velocity_x(i,j)*dt)/(2*dx))*((1-sign(
            velocity_x(i,j)))*(wold(i+1,j)-wold(i,j))+(1+sign(velocity_x(
            i,j)))*(wold(i,j)-wold(i-1,j)))...
        %                     -  ((velocity_y(i,j)*dt)/(2*dy))*((1-sign(
            velocity_y(i,j)))*(wold(i,j+1)-wold(i,j))+(1+sign(velocity_y(
            i,j)))*(wold(i,j)-wold(i,j-1)))...
        %                     + (dt/(Re*dx^2))*(wold(i+1,j)+wold(i-1,j)
            -2*wold(i,j)) + (dt/(Re*dy^2))*(wold(i,j+1)+wold(i,j-1)-2*
            wold(i,j));

        %explicit conservative
        % w(i,j)= wold(i,j) - (dt/(2*dx))*((velocity_x(i+1,j)*wold(i+1,j
            ))-(wold(i-1,j)*velocity_x(i-1,j)))...
        %                   -(dt/(2*dy))*((velocity_y(i,j+1)*wold(i,j+1)
            )-(wold(i,j-1)*velocity_y(i,j-1)))...
        %                   + (dt/(Re*dx^2))*(wold(i+1,j)+wold(i-1,j)-2*
            wold(i,j)) + (dt/(Re*dy^2))*(wold(i,j+1)+wold(i,j-1)-2*wold(i
            ,j));

        % same but faster because there is no wold about 15% faster
        w(i,j)= w(i,j) - (dt/(2*dx))*((velocity_x(i+1,j)*w(i+1,j))-(w(i
            -1,j)*velocity_x(i-1,j)))...
                    -(dt/(2*dy))*((velocity_y(i,j+1)*w(i,j+1))-(w(i,j
                        -1)*velocity_y(i,j-1)))...
                    + (dt/(Re*dx^2))*(w(i+1,j)+w(i-1,j)-2*w(i,j)) + (
                        dt/(Re*dy^2))*(w(i,j+1)+w(i,j-1)-2*w(i,j));
    end
end
%%  calculate psi
D1=1;
k1=0;
while D1 >= tolpsi && k1<kmax
    k1=k1+1;
    psiold=psi;
    for i=2:N-1
        for j=2:M-1
            psi(i,j) = f*(1/(2*(1+Beta2)))*(psi(i+1,j)+psi(i-1,j)+Beta2
                *(psi(i,j-1)+psi(i,j+1))+(dx^2)*w(i,j))+(1-f)*psiold(i,j)
                ; %SOR
        end
    end
    D1=max(max(abs(psi-psiold)));
end
%calculation of the velocity
for i=2:N-1
    for j=2:M-1
        velocity_y(i,j)=-(psi(i+1,j)-psi(i-1,j))/(2*dx);
        velocity_x(i,j)=(psi(i,j+1)-psi(i,j-1))/(2*dy);
    end
end
```

```
    %calculation of the vorticity on walls
    for i=1:N
        w(i,M)=(1/(2*dy^2))*(-8*psi(i,M-1)+psi(i,M-2)-6*dy*Un); % North wall
        w(i,1)=(1/(2*dy^2))*(-8*psi(i,2)+psi(i,3)+6*dy*Us); % South wall
    end
    for i=1:M
        w(N,i)=(1/(2*dx^2))*(-8*psi(N-1,i)+psi(N-2,i)+6*dx*Ve); % East wall
        w(1,i)=(1/(2*dx^2))*(-8*psi(2,i)+psi(3,i)-6*dx*Vw); % West wall
    end

    if(mod(k,100)==0)
          fprintf('%5d %5d %5e %5d \n',k,k1,Err,max(max(psi)));
    end
    Err=max(max(abs(psi-psi_old)));

end
ttot= toc;
```

References

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[3] Beckers, J. M. (1992). Analytical Linear Numerical Stability Conditions for an Anisotropic Three-Dimensional Advection-Diffusion Equation. SIAM Journal on Numerical Analysis, 29(3), 701–713. http://www.jstor.org/stable/2158274